

Survey High-Frequency Data Quality Checks

Setup & Introduction Guide

Automated enumerator monitoring and indicator quality control for food security surveys

Standards	World Bank DIME iehfc · SurveyCTO HFC framework
Mode	Local installation — data never leaves your machine
Formats	CSV · Excel · Stata (.dta) · SPSS (.sav)
Requires	Python 3.9 or higher

Introduction

What is Survey HFC?

Survey High-Frequency Data Quality Checks (HFC) is a locally-run dashboard for automated data quality monitoring on food security and household survey data. Upload your dataset daily and get a structured view of enumerator behaviour, flag rates, duplicate records, and actionable field feedback — before errors compound.

The tool runs ten indicator modules against your survey data and surfaces issues across four review areas: **sequential checks** that catch structural impossibilities before other checks run; **indicator-level checks** that flag implausible or inconsistent values within each module; **cross-indicator checks** that detect logical contradictions across indicators using FEWS NET illogicality pairs; and **enumerator monitoring** that surfaces patterns suggesting coaching, rushing, or data fabrication.

Key Features

Ten Indicator Modules

Demographics, FCS, HDDS, rCSI, Housing, LCS, Food Expenditure, Non-food Expenditure (1-month), Non-food Expenditure (6-month), and Interview Timing — all configurable from the app UI without touching any files.

Enumerator Monitoring

Per-enumerator flag rates vs. overall average, indicator means comparison with 1.5 SD outlier highlighting, daily submission heatmap by enumerator, and missing data rate by indicator and enumerator.

Survey Progress Tracking

Cumulative and daily submission charts with a target line, pace projections (average surveys per day, estimated days to completion), and an area completion table with status indicators.

Cross-Indicator Logic Checks

Detects contradictions across indicators using FEWS NET illogicality pairs — combinations that cannot coexist under standard food security frameworks, such as Poor FCS with zero coping strategies.

Three Export Formats

Colour-coded Excel error log (one sheet per indicator), self-contained HTML manager report (print to PDF via browser), and per-enumerator feedback Excel (one sheet per enumerator with only their flagged records).

Local & Offline Operation

The application runs entirely on the user's machine. No survey data, results, or any other information is transmitted to any external server at any point.

Dashboard Outputs

Tab / Output	Description
Survey Status	Cumulative and daily submission counts, progress vs. target, pace projections, priority action items, enumerator and area completion tables

Tab / Output	Description
Enumerator Behavior	Flag rates per enumerator, indicator means comparison table with outlier highlighting, daily submission heatmap
Data Quality	Duplicate HHID detection, flag rates by indicator, FCS and rCSI distributions, FEWS NET illogicality matrix, missing data heatmap
Flag Details	Full flagged record tables per indicator with plain-language narrative explanations
HTML Report	Self-contained manager report — print to PDF via browser, no internet required
Excel Error Log	Colour-coded workbook with one sheet per indicator, ready for field team review
Enumerator Feedback Excel	One sheet per enumerator showing only their flagged records, with a summary sheet

How to Use the Tool

1 Install the application

Follow the platform-specific setup instructions in this guide. Installation is required only once; subsequent launches use the provided one-click launcher script.

2 Launch and upload your data

Start the app and upload your survey file (CSV, Excel, Stata .dta, or SPSS .sav). The tool auto-detects context columns — HHID, enumerator, area, and date.

3 Select indicators and adjust thresholds

Enable the indicator modules present in your questionnaire. All thresholds (FCS cut-offs, timing limits, expenditure ceilings, minimum price) can be adjusted in the app UI without editing any files.

4 Run Checks and review results

Click Run Checks. Navigate the tabs to review flag rates by indicator and enumerator, logical contradictions, duplicates, and submission progress.

5 Download field team reports

Generate the Excel error log and per-enumerator feedback files. Share with supervisors or enumerators for daily field review.

Data Privacy Statement

Survey HFC processes all data locally on the user's machine. No survey data, results, or any other information is transmitted to any external server, cloud service, or third party at any point during use.

Local Installation Guide

What You Need

Confirm you have the project folder and place it anywhere on your computer (for example, your Desktop). The folder should contain:

File / Folder	Purpose
app.py	The application
requirements.txt	List of software dependencies
config/	Threshold and column definition files
hfc/	Indicator check modules
Run HFC Checks.bat	One-click launcher for Windows
Run HFC Checks.command	One-click launcher for macOS

macOS Instructions

Step 1 Check Python is installed

Open **Terminal** (+ Space, type *Terminal*, press Enter) and run the command above. You should see **Python 3.9.x** or higher.

If Python is not found: download the latest 3.11.x installer from python.org/downloads and follow the prompts.

```
python3 --version
```

Step 2 Navigate to the app folder

Run `ls` to confirm `app.py` and `requirements.txt` are listed.

```
cd ~/Desktop/survey-hfc
```

Step 3 One-click launch (recommended)

Double-click **Run HFC Checks.command**. On first run, macOS may block the file — right-click **Open** to allow execution. You only need to do this once. The script creates a virtual environment, installs all dependencies, and opens the app in your browser at `http://localhost:8501`.

Step 4 Manual launch (alternative) — Create virtual environment

Creates an isolated Python environment. First time only.

```
python3 -m venv venv
```

Step 5 Activate the virtual environment

The terminal prompt will show (venv) when active.

```
source venv/bin/activate
```

Step 6 Install dependencies

Downloads all required packages. This may take 1–2 minutes on first run.

```
pip install -r requirements.txt
```

Step 7 Launch the application

A browser window will open at <http://localhost:8501>. If it does not open automatically, paste that address into any browser.

```
streamlit run app.py
```

Returning users — quick launch

```
cd ~/Desktop/survey-hfc
source venv/bin/activate
streamlit run app.py
```

To **stop the app**, press `Ctrl + C` in Terminal.

Windows Instructions

Step 1 Check Python is installed

Open **Command Prompt** (Win + R, type *cmd*, press Enter). You should see **Python 3.9.x** or higher.

If Python is not found: download the 3.11.x 64-bit installer from python.org/downloads/windows. On the first screen, tick "**Add Python to PATH**" before clicking Install Now.

```
python --version
```

Step 2 Navigate to the app folder

Run `dir` to confirm `app.py` and `requirements.txt` are listed.

```
cd %USERPROFILE%\Desktop\survey-hfc
```

Step 3 One-click launch (recommended)

Double-click **Run HFC Checks.bat**. If Windows Defender SmartScreen blocks it, click **More info Run anyway**. The script creates a virtual environment on first run, installs all dependencies, and opens the app in your browser at `http://localhost:8501`.

Step 4 Manual launch (alternative) — Create virtual environment

```
python -m venv venv
```

Step 5 Activate the virtual environment

The prompt will show `(venv)`.

```
venv\Scripts\activate
```

*If activation is blocked by an execution policy error, open PowerShell as Administrator and run:
`Set-ExecutionPolicy RemoteSigned -Scope CurrentUser` Then return to Command Prompt and try again.*

Step 6 Install dependencies

```
pip install -r requirements.txt
```

Step 7 Launch the application

A browser window will open at `http://localhost:8501`.

```
streamlit run app.py
```

Returning users — quick launch

```
cd %USERPROFILE%\Desktop\survey-hfc
venv\Scripts\activate
streamlit run app.py
```

To **stop the app**, press `Ctrl + C` in Command Prompt.

Column Naming Reference

Column names must match the VAM convention exactly (case-sensitive). The tool auto-detects context columns and matches indicator columns by name.

Context Columns (auto-detected)

Purpose	Accepted column names
Household ID	HHID hhid _uuid uuid instanceID ID
Enumerator	EnuName enumerator_name enum_name EnuID enumerator — or any column containing <i>enumerat</i>
Supervisor	EnuSupervisorName supervisor_name supervisor — or any column containing <i>supervis</i>
Admin area	ID02 ID01 admin2 admin1 district region woreda county
Survey date	SvyDate today _submission_time SubmissionDate date

Food Consumption Score (FCS)

Values = days of consumption in the past 7 days (0–7). FCS = sum of weighted values (max 112).

Classification: Poor ≤21 · Borderline 21.5–35 · Acceptable >35.

Column	Food group	Weight
FCSStap	Cereals and starchy staples	×2
FCSPu lse	Pulses, legumes and nuts	×3
FCSDairy	Dairy products	×4
FCSPr	Meat, fish and eggs	×4
FCSVeg	Vegetables	×1
FCSFruit	Fruits	×1
FCSFat	Oil and fats	×0.5
FCSSugar	Sugar and sweets	×0.5

Reduced Coping Strategies Index (rCSI)

Values = days the strategy was used in the past 7 days (0–7). rCSI = sum of weighted values (max 56).

Crisis threshold: ≥19.

Column	Strategy	Weight
rCSILessqlty	Relied on less preferred or cheaper food	×1
rCSIBorrow	Borrowed food or relied on help from others	×2

Column	Strategy	Weight
rCSIMealNb	Reduced number of meals per day	×1
rCSIMealSize	Restricted portion sizes at mealtimes	×1
rCSIMealAdult	Adults restricted intake so children could eat	×3

Timing

Column	Description
start	Interview start timestamp — timezone-aware ISO 8601 (ODK / SurveyCTO device time)
end	Interview end timestamp — must be after start

Livelihood Coping Strategies (LCS)

Value	Meaning
10	Strategy applied
20	Not needed (household has enough food)
30	Exhausted — no longer an option
9999	Not applicable to this household

Expenditure Column Pattern

Columns follow the pattern {Category}{Source}, for example HHExpFCer_Purch_MN_7D.

Recall period	Sources	Food category prefixes
Food — 7-day	_Purch_MN_7D _GiftAid_MN_7D _OwnProd_MN_7D	HHExpFCer HHExpFTub HHExpFPulse HHExpFVeg HHExpFFruit HHExpFMeat HHExpFFish HHExpFDairy HHExpFEgg HHExpFOil HHExpFSugar HHExpFCond HHExpFBev HHExpFOther
Non-food — 1-month	_Purch_MN_1M _GiftAid_MN_1M	HHExpNFHyg HHExpNFTrans HHExpNFFuel HHExpNFWat HHExpNFComm HHExpNFMed HHExpNFEduc HHExpNFOther
Non-food — 6-month	_Purch_MN_6M _GiftAid_MN_6M	HHExpNFCloth HHExpNFRent HHExpNFDurable HHExpNFCerem HHExpNFDebt

Configuration Reference

All thresholds can be adjusted in the app UI without touching any files. To make changes permanent for a specific deployment, edit the YAML files in `config/configurable/`.

FCS — `config/configurable/fcs.yaml`

Parameter	Default	Description
<code>low_fcs_threshold</code>	10	Flag FCS below this value
<code>high_fcs_threshold</code>	100	Flag FCS above this value
<code>low_staple_threshold</code>	2	Flag FCSStap at or below this value
<code>fcg_poor_threshold</code>	21	FCS Poor classification cut-off
<code>fcg_borderline_threshold</code>	35	FCS Borderline classification cut-off

rCSI — `config/configurable/rcsi.yaml`

Parameter	Default	Description
<code>high_rcsi_with_acceptable_fcg</code>	18	Flag high rCSI combined with Acceptable FCS category

Timing — `config/configurable/timing.yaml`

Parameter	Default	Description
<code>short_duration_min</code>	20	Flag interviews shorter than this many minutes
<code>long_duration_min</code>	120	Flag interviews longer than this many minutes
<code>utc_offset_hours</code>	0	Local timezone offset from UTC
<code>abnormal_early_morning_end</code>	7	Flag interviews starting before this hour (24h)
<code>abnormal_evening_start</code>	19	Flag interviews starting after this hour (24h)

Expenditure — `config/configurable/hhexp.yaml`

Parameter	Default	Description
<code>min_item_price</code>	0	Minimum meaningful price in local currency. Any non-zero item below this value is flagged as implausibly small. Set to the price of a basic local purchase (e.g. bread, bus token). Leave at 0 to disable.
<code>max_single_item_food_7d</code>	1,000,000	Flag food items above this value
<code>max_single_item_nonfood_1m</code>	1,000,000	Flag short-term non-food items above this value

Parameter	Default	Description
max_single_item_nonfood_6m	5,000,000	Flag long-term non-food items above this value
flag_zero_total_food	true	Flag households where all food expenditure is zero

Demographics — config/configurable/demo.yaml

Parameter	Default	Description
high_hhsize_threshold	30	Flag households larger than this size

Troubleshooting

Problem	Likely Cause	Solution
python3: command not found	Python is not installed	Follow Step 1 for your operating system
pip: command not found	pip not on PATH	Use <code>python3 -m pip install -r requirements.txt</code>
ModuleNotFoundError: No module named 'streamlit'	Virtual environment not active	Run the activate command before launching
No flags generated after upload	Column names do not match convention	Check names against the column reference — matching is case-sensitive
Timing module produces no output	Timestamps not timezone-aware	Ensure start and end are in ODK/SurveyCTO default ISO 8601 format
Browser does not open automatically	Streamlit cannot detect browser	Navigate manually to <code>http://localhost:8501</code>
Port 8501 already in use	Another Streamlit instance is running	Stop it with <code>Ctrl + C</code> , or use <code>--server.port 8502</code>
Windows: activate execution policy error	PowerShell security setting	Run <code>Set-ExecutionPolicy RemoteSigned -Scope CurrentUser</code> in PowerShell as Administrator
macOS: launcher blocked	Gatekeeper security setting	Right-click Open Open
Stata file fails to load	Duplicate value labels	The tool will retry automatically with numeric codes and show an info message

Frequently Asked Questions

Is my data safe?

Yes. The application runs entirely on your local machine. No survey data, results, or any other information is transmitted to any external server at any point.

Can I run this without an internet connection?

Yes. Once dependencies are installed the app runs fully offline. An internet connection is only needed the first time, to install packages.

Can I use data that does not follow the VAM column naming?

The indicator modules require exact column name matches. You can rename columns in your data export, or adjust the column lists in `config/standard/.yaml` to match your own naming convention.

How do I add a new country or context?

Adjust the thresholds in `config/configurable/` for the specific context (currency scale, working hours, household size norms). No code changes are needed.

What is the minimum meaningful price parameter?

Set `min_item_price` in `config/configurable/hhexp.yaml` to the price of the cheapest plausible local purchase (e.g. a loaf of bread or a bus token). Any expenditure item recorded as non-zero but below this floor will be flagged as a likely digit-drop entry error. Leave at 0 to disable.

Where are downloaded reports saved?

Reports are saved to your browser's default download folder.

How do I update the app when a new version is available?

Replace the project files with the new version. Re-run `pip install -r requirements.txt` with the virtual environment active to update any changed dependencies.

Survey HFC · Aligned with iehfc (World Bank DIME) and SurveyCTO HFC frameworks